SYSTEM WALKTHROUGH

Paper Attribution and Connection Intelligence

How the AC bibliometric system explains papers, tracks relationships, and prepares source-grounded knowledge for cross-SDL reuse.

01 ATTRIBUTION

02 CONNECTIONS

03 KNOWLEDGE

Classification is the first decision, not the product

1

CLASSIFICATION

Which relevance class is plausible?

2

PAPER ATTRIBUTION

Which AC people, institutions, funding, citations, and graph paths support it?

3

CONNECTION INTELLIGENCE

Which authors, institutions, topics, and methods are connected across the corpus?

4

KNOWLEDGE REUSE

Which source-grounded methods and lessons can transfer to another SDL?

DURABLE OUTPUT

A provenance-preserving research graph in which every node, edge, label, and generated claim can be traced back to papers and source text.

Paper attribution resolves five relationships around every paper



Human review records the final decision and locks it against later automated overrides.

What the system stores for one attributed paper

OUTPUT FIELD	WHAT IT CONTAINS	WHY IT MATTERS
paper identity	DOI, title, year, venue, source provenance	prevents duplicate or ambiguous records
classification	label, confidence, model version, reasons	supports filtering and review priority
matched people	person ID, name, match type: exact / alias / initials / fuzzy / AC publication	distinguishes identity evidence from topic similarity
matched institutions	canonical institution ID and partner status	supports partner and collaboration reporting
scientific evidence	topics, methods, entities, full-text evidence, chunk IDs	keeps the scientific basis source-addressable
graph explanation	direct AC authors, second-degree paths, AC-core citations, cited-by-corpus	makes indirect attribution inspectable
human review	human label, reviewer, notes, timestamp / lock state	preserves accountable decision authority

Seven evidence signals feed label-specific gates

A	I	T	F	N	Q	E
author identity	institution	topic	full text	network	citations	entities

$$C = .35A + .15I + .25T + .20F + .10N + .10Q + .05E - .15D$$

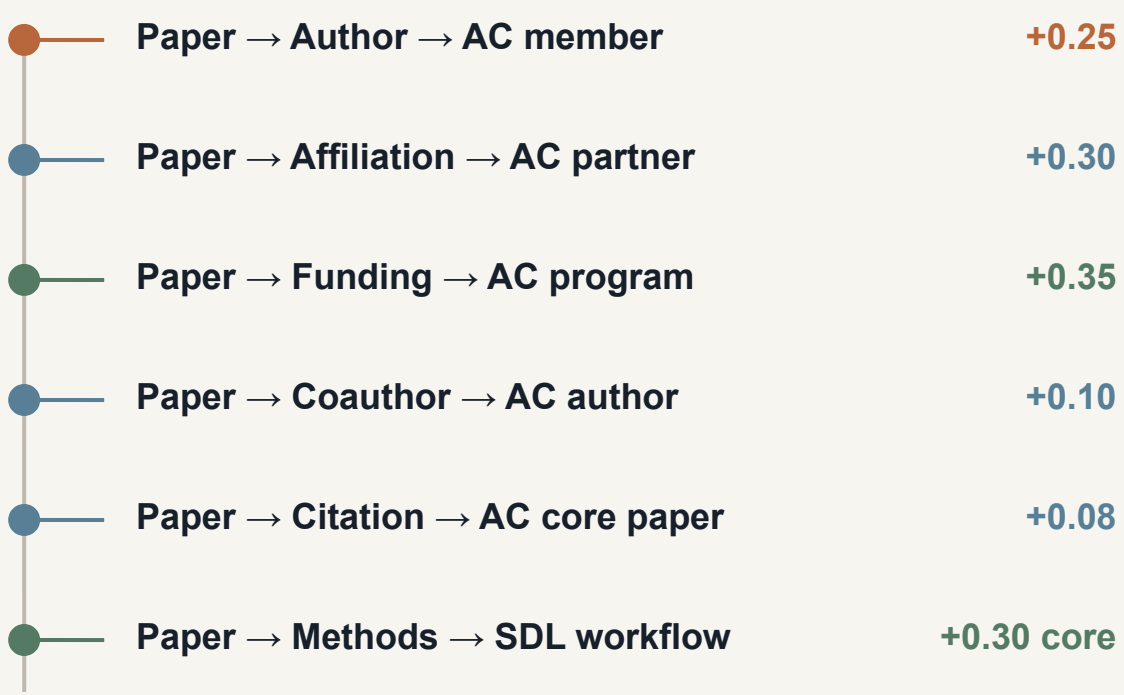
D = disqualifying evidence count. Metadata-only mode redistributes weight toward A, I, and T.

IMPORTANT

Confidence ranks cases. Necessary conditions assign labels. A high score cannot bypass the evidence required for a high-value attribution.

CORE	direct / gold AC evidence + core SDL evidence
RELATED	strong identity, funding, or AC-core citation support
ADJACENT	strong scientific relevance without sufficient attribution
UNCERTAIN	partial or conflicting evidence requiring review
EXCLUDE	no necessary condition survives

Attribution is an evidence tree, not a flat score



S(AET) = clamp(Σ positive path weights − Σ negative path weights, 0, 1)

EACH PATH RETAINS

- path ID and evidence type
- polarity and weight
- core vs supporting centrality
- source section and chunk ID
- generation and review context

Negative evidence remains visible; it is not discarded after scoring.

Every graph claim can be opened as a path

DIRECT

Paper → AC author

The paper has a verified AC member among its authors.

SECOND DEGREE

Paper → external author → AC author

The path records the two authors, shared paper IDs, and a path strength.

CITATION

Paper → cites → AC core paper

The cited AC-core title and DOI remain attached to the signal.

GRAPH EXPLANATION OUTPUT

direct_ac_authors[]	verified names
second_degree_paths[]	via author · AC author · shared papers · strength
citation_signals[]	AC-core paper · title · DOI
cited_by_corpus[]	reverse corpus citations
summary	deterministic explanation text

$$N(p) = \max_a \text{PPRAC}(a) \times [1 + .25 \cdot \max(0, \text{cnorm}(a) - 1)]$$

$$Q(p) = \min(1, 0.5 \times |\text{ unique AC-core citations } |)$$

Raw co-publication counts are normalized before they become relationship strength

R(u,v) = shared papers(u,v) / min(paper count(u), paper count(v))

This asks how much of the smaller publication portfolio is explained by the collaboration.

EXAMPLE	SHARED	PAPERS (u,v)	RAW IMPRESSION	NORMALIZED R
A	3	4 and 12	moderate tie	**0.75**
B	5	30 and 20	larger raw count	**0.25**

Example A is the more concentrated relationship even though it has fewer shared papers.

ADDITIONAL STRUCTURAL SIGNALS

PPR toward AC

Proximity to AC-seeded authors through the weighted coauthor graph.

Evans–Chen excess centrality

Highlights authors whose collaborative reach is larger than degree alone would predict.

Bridge score

B(a) = min[1, PPRAC(a) × max(cnorm(a),0) × SDL diversity(a)]

AC can rank the same network for different questions

$$I_{\square}(a) = \Sigma_{\square} w_{\square,\square} \times x_{\square}(a)$$

PROFILE	PRIMARY QUESTION	NON-ZERO WEIGHTS
AC attribution	Who is structurally closest to AC work?	AC proximity .55 · relationship .20 · citation .15 · recency .10
Bridge builders	Who connects AC-adjacent communities across SDL contexts?	AC proximity .35 · excess centrality .30 · SDL diversity .25 · relationship .10
Institution hubs	Who sustains institution-spanning collaboration?	relationship .30 · excess centrality .25 · SDL diversity .20 · AC proximity .15 · citation .10
Emerging connectors	Which newer authors show early bridge potential?	recency .30 · excess centrality .25 · SDL diversity .20 · AC proximity .15 · relationship .10
Topic diffusion	Who helps methods and topics spread between SDL areas?	SDL diversity .35 · relationship .20 · excess centrality .20 · AC proximity .15 · recency .10

The UI exposes both the profile weights and each dimension’s contribution; profile changes are therefore explainable rather than opaque.

The coauthor view exposes bridge authors and the evidence behind each edge

WHAT AC CAN INSPECT

Question-specific ranking

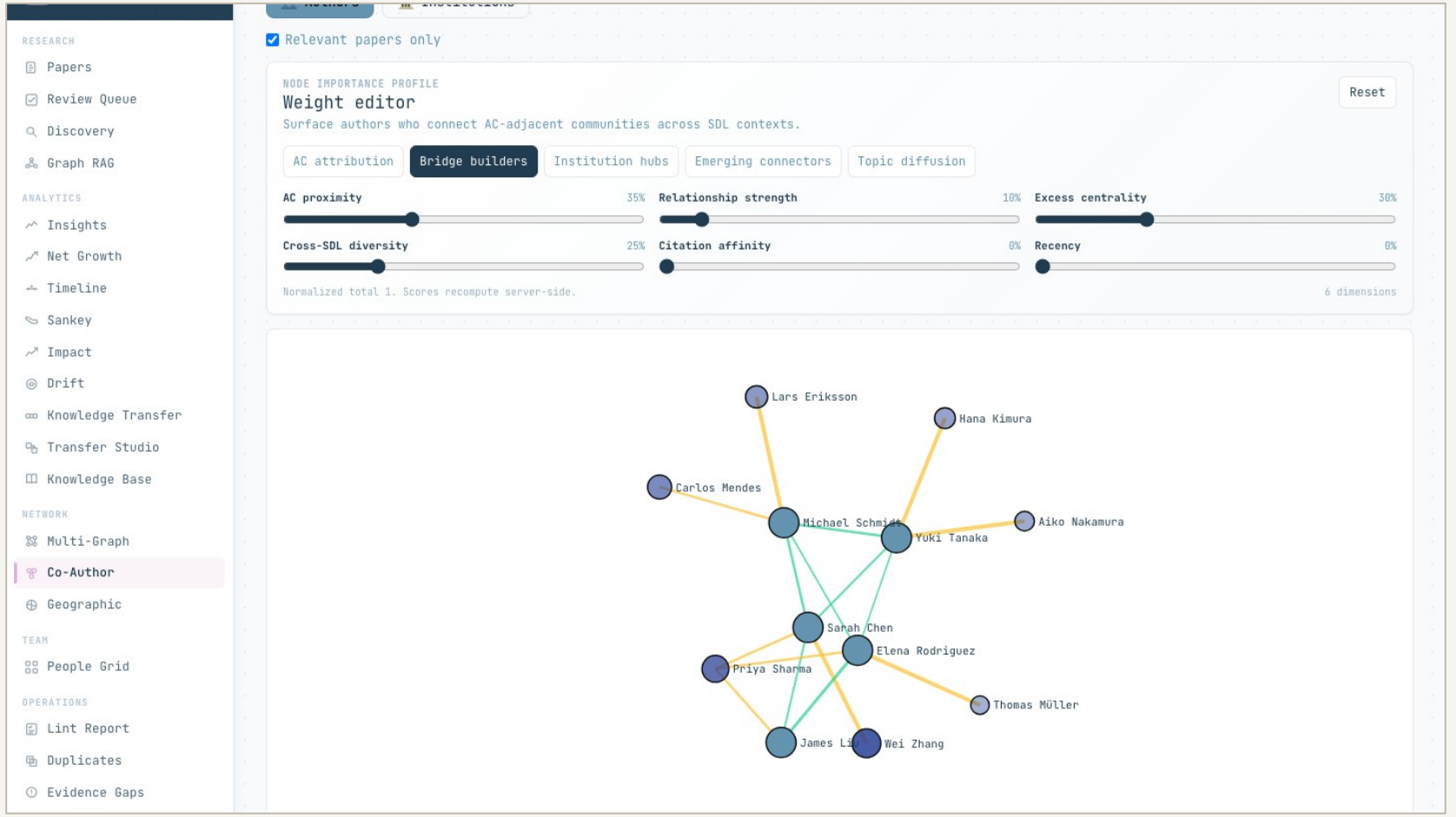
Switch between AC attribution, bridge builders, institution hubs, emerging connectors, and topic diffusion.

Author-level explanation

Paper count, collaborators, PPR, cnorm, importance components, and a human-readable explanation.

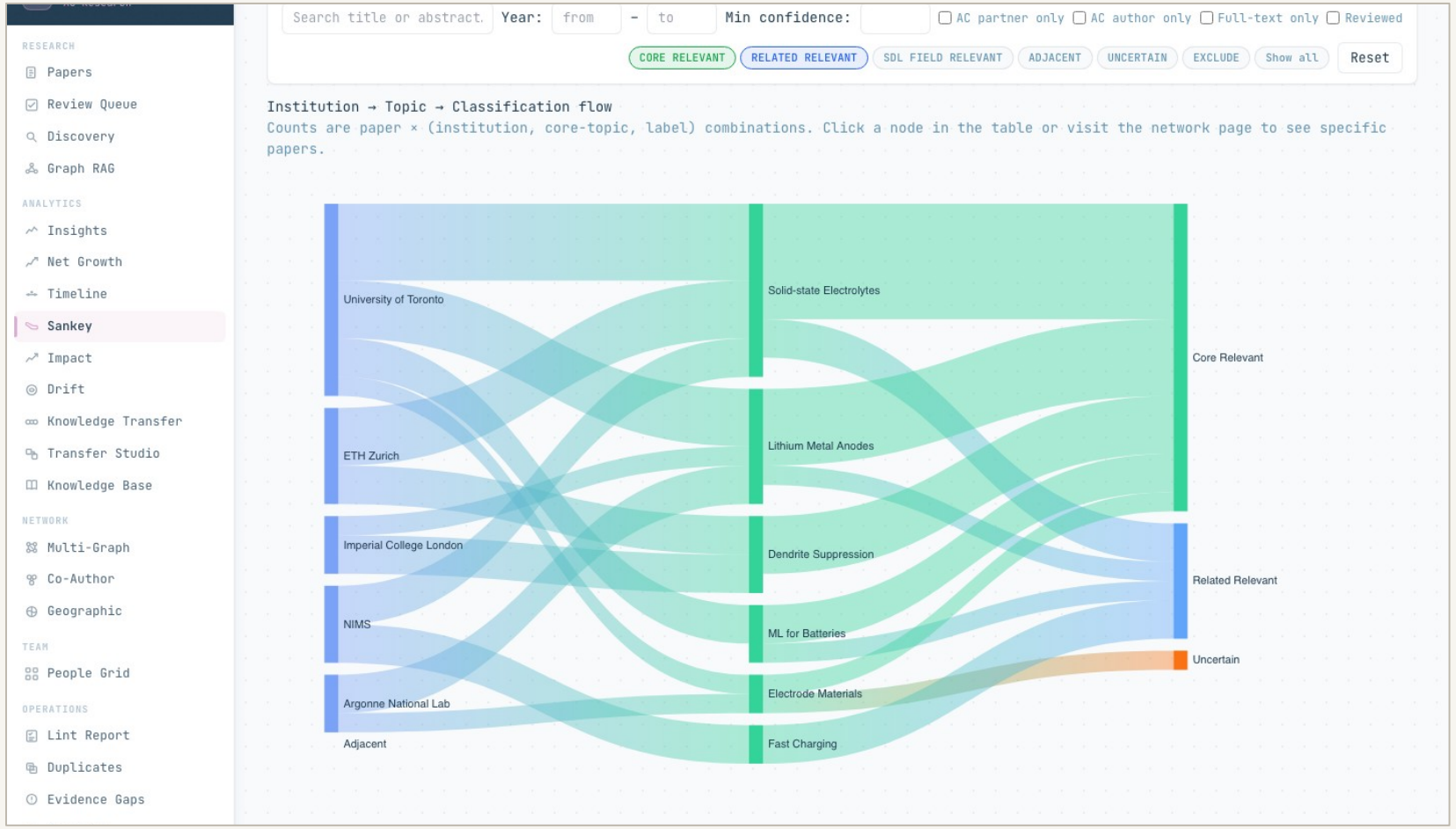
Edge provenance

Joint-paper IDs, normalized relationship strength, and whether an edge involves AC.



Interface captured from the current frontend with mock data; values are illustrative.

Institution connections can be traced from joint papers to topics and labels



Interface captured from the current frontend with mock data; values are illustrative.

THREE LINKED QUESTIONS

INSTITUTION

Which organizations appear on the attributed papers?

TOPIC

Which scientific themes connect those organizations?

LABEL

How does the evidence resolve into attribution classes?

INSTITUTION HUB SCORE

H = .4 degree + .4 betweenness + .1 partner reach + .1 international reach

The multi-entity graph joins people, institutions, papers, topics, methods, funders, and AC core

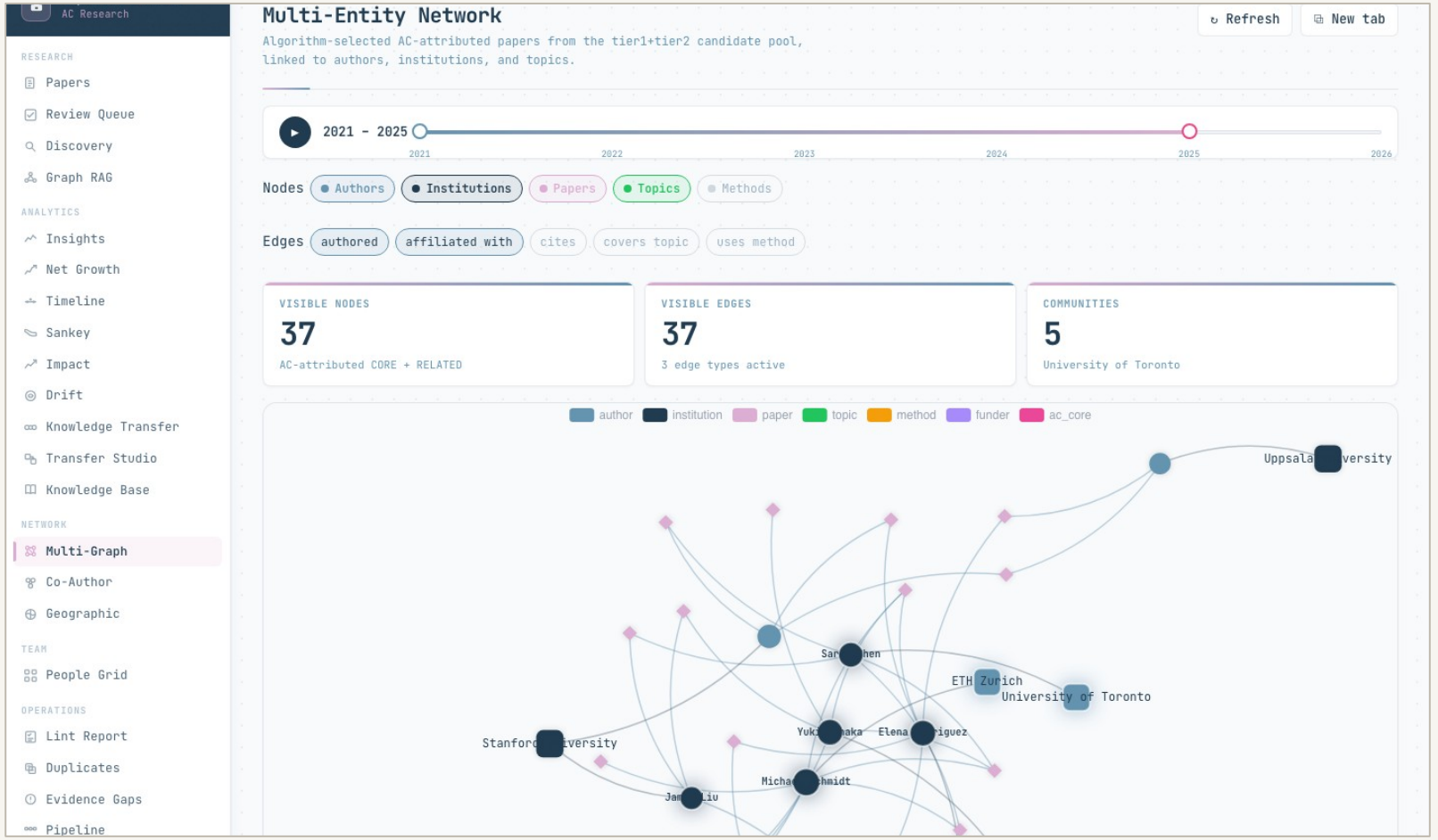
NODE TYPES

authors
institutions
papers
topics
methods
funders
AC core

EDGE PREDICATES

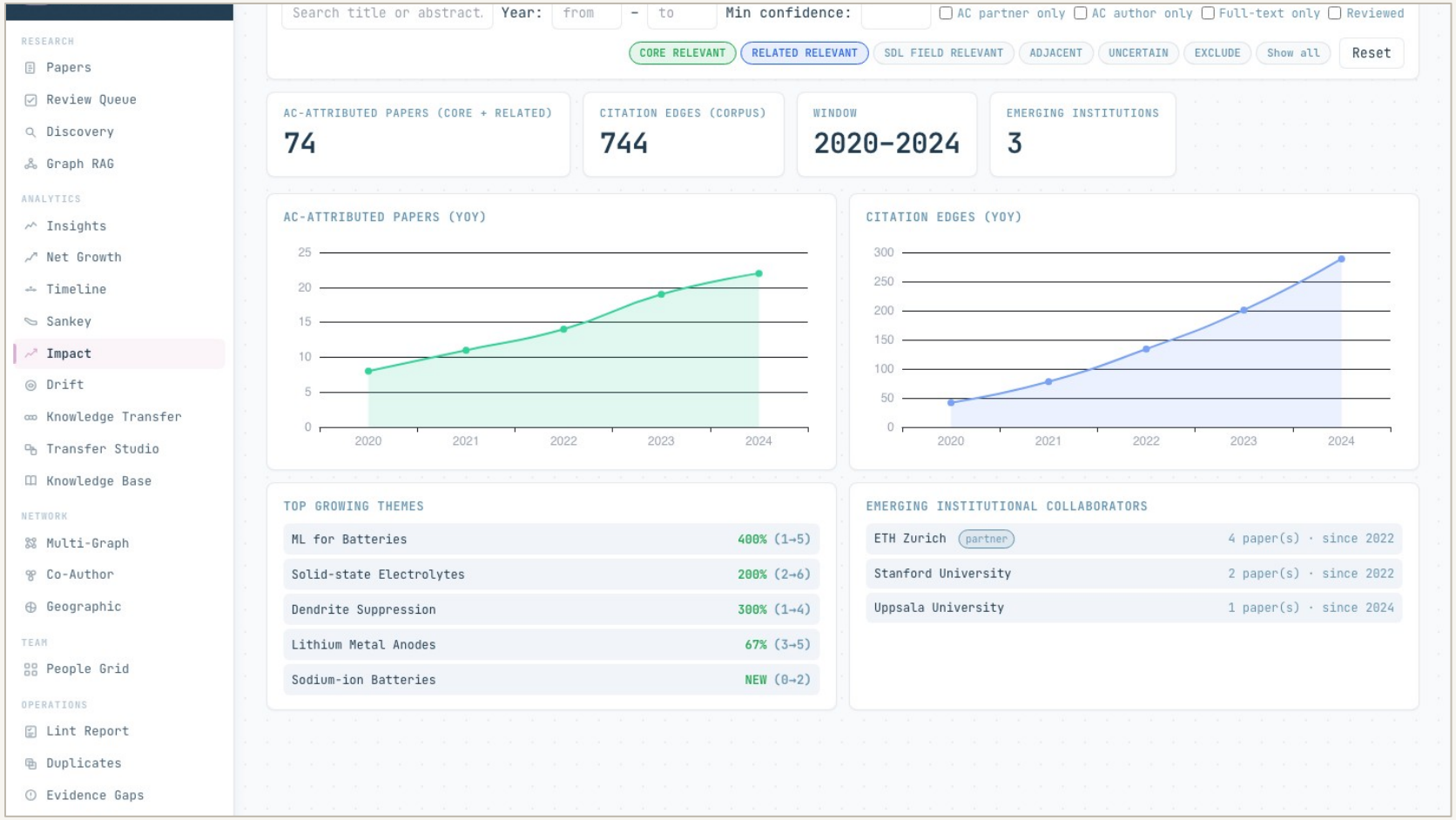
authored
affiliated with
belongs to topic
uses method
cites / cites AC core
institution collaboration

Center any entity, set a hop limit, choose edge types, and move the time window without changing the underlying provenance.



Interface captured from the current frontend with mock data; values are illustrative.

AC can monitor growth, emerging collaborators, themes, and citation activity



Interface captured from the current frontend with mock data; values are illustrative.

METRICS ARE SQL-DERIVED

Attributed papers

CORE + RELATED by year

Citation activity

corpus citation edges by year

Emerging institutions

first year + paper count + partner status

Emerging authors

first year + attributed-paper activity

Growing themes

recent versus earlier topic counts

Drift events

year-to-year topic-vector similarity

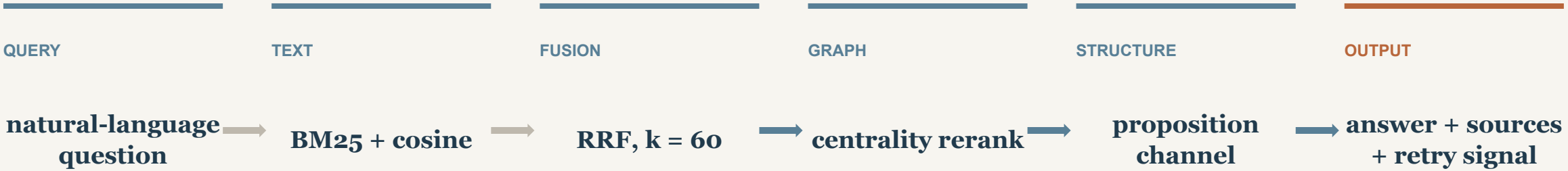
LLMs do not calculate these numbers.

Connection intelligence supports concrete operating questions

OPERATING QUESTION	SYSTEM EVIDENCE	POSSIBLE AC ACTION
Which papers are attributable to AC, and why?	paper record · AET · graph explanation	review, publish, or correct the attribution
Who bridges AC to an external SDL community?	PPR · excess centrality · SDL diversity · joint papers	prioritize outreach or collaboration review
Which institutions are emerging collaborators?	first-year activity · institution edges · hub score · geography	support partnership development
Which topics or methods move across communities?	Sankey · multi-graph · topic diffusion profile	identify translation opportunities
Where are the evidence and ingestion blind spots?	evidence gaps · duplicate candidates · missed-paper groups · drift	improve dataset coverage and review policy
Which claims can go into a stakeholder report?	SQL metric triples with source links	produce auditable reporting without invented numbers

The system turns bibliometric records into a decision surface for curation, partnership tracking, and scientific translation.

Retrieval answers questions with graph context, propositions, and source chunks



$$RRF(d) = 1 / [60 + rankBM25(d)] + 1 / [60 + rankcosine(d)]$$

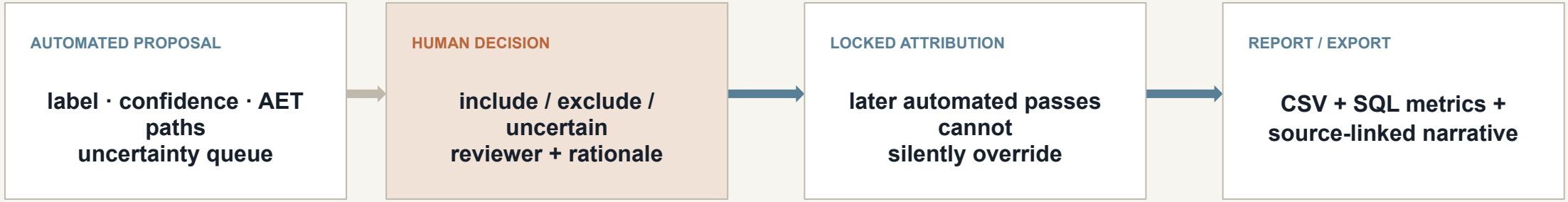
$$score'(d) = RRF(d) \times [1 + 0.3 \times graph\ importance(d)]$$

EVERY RETURNED SOURCE INCLUDES

paper ID and chunk ID
section type and title
BM25, cosine, fused, and graph scores
graph-path context when available
source precision and missing-evidence signals

Structured questions about methods or funding can use proposition-first retrieval instead of relying only on free text.

Human review and reporting preserve authority



SAFE ROLE FOR CLAUDE

- adjudicate only an uncertainty queue
- summarize retrieved evidence
- draft reviewer rationale
- rewrite reports from supplied metric triples

AUTHORITY BOUNDARY

Claude may not create papers, graph edges, citations, measurements, results, or numerical claims. Human-confirmed decisions remain locked.

Attributed papers become source-grounded cross-SDL transfer candidates



Transfer score = match strength + novelty + transferability + evidence − risk penalty

SOURCE CONTRACT

Every generated claim carries a source tag such as [src:paper_id]. Same-domain matches are excluded. Human review precedes acceptance.

Classification makes the corpus searchable; attribution makes the extracted knowledge defensible.

Current capability, optional layers, and experimental work remain distinct

STATUS	CAPABILITIES	AUTHORITY / LIMIT
CURRENT	paper attribution record · AET evidence paths · deterministic graph explanations · coauthor / institution / multi-entity networks · Sankey / impact / drift · source-linked retrieval · TKU and transfer services	implemented paths in the current repository; outputs remain evidence-linked
OPTIONAL	Claude uncertainty judge · Claude report rewriter · learned classifier using 14 engineered features	default-off or conditional; bounded by retrieved evidence and high-margin rules
OFFLINE BASELINE	ten leakage-safe structural graph features + balanced logistic regression + leave-one-out evaluation	evaluation artifact only; no production authority
FUTURE EXPERIMENT	PyTorch Geometric / DGL model training, saved weights, inference service, production classifier integration	not implemented as a production GNN path

Do not describe graph export or the offline structural baseline as a deployed GNN classifier.

Decisions for a shared bibliometric program

The first release should define not only which papers enter the dataset, but which relationships and evidence paths the consortium agrees to trust.

01 **CORPUS**

Which bibliometric sources, date range, and gold papers define the trusted starting set?

02 **ATTRIBUTION POLICY**

Which evidence paths are sufficient for CORE, RELATED, and organization-level reporting?

03 **CONNECTION METRICS**

Which profiles should AC use for attribution, bridge discovery, institutions, and topic diffusion?

04 **REVIEW & MODEL BUDGET**

Which uncertainty bands require humans, and where should Claude credits reduce review work?